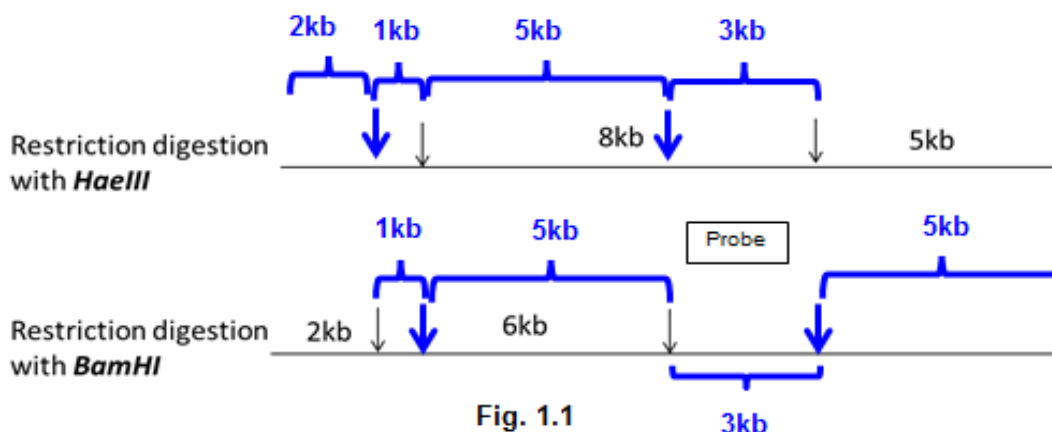




## H2 BIOLOGY PAPER 3 **ANSWER SCHEME**

### Question 1

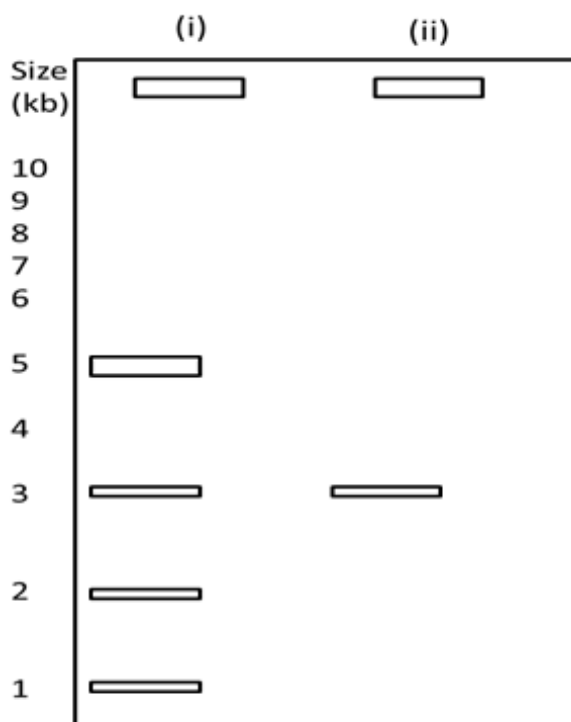
Restriction digestion, gel electrophoresis and Southern Blotting are important molecular techniques used in disease detection and in paternity testing.



A restriction digestion using both *HaeIII* and *BamHI* was performed. On the diagram below, show the band pattern of the restriction digestion,

- after gel electrophoresis and staining with DNA binding dye. [1]
- after gel electrophoresis and nucleic acid hybridization with the probe shown in **Fig. 1.1**. [1]

Assume that complete restriction digestion by both restriction enzymes has taken place.



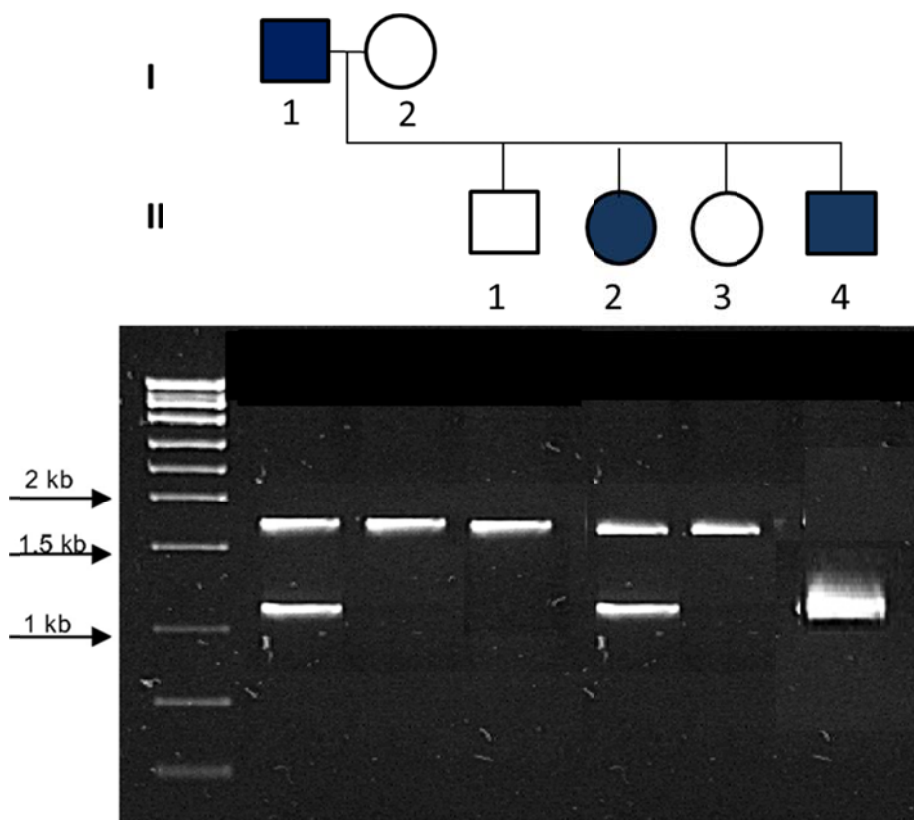
(iii) Explain the difference in the band pattern drawn for (i) and (ii). [2]

- DNA binding dye **intercalates/binds to all DNA fragments**, and hence under UV light, **all DNA fragments will be observed**.
- Probe however only **hybridize/anneal to fragments with complementary sequence**, hence only specific fragments hybridized with the probe will be observed during autoradiography.

Various molecular techniques can be used in disease detection. One such disease is the Marfan syndrome. Marfan syndrome is an inherited condition of the connective tissue caused by mutation of the *FBN1* gene.

In the detection of Marfan syndrome, the mature mRNA of *FBN1* gene is detected using reverse transcription polymerase chain reaction (RT-PCR). In RT-PCR, mature mRNA of *FBN1* gene is first reverse transcribed to its complementary DNA using reverse transcriptase. The products are processed before traditional PCR is applied to amplify the cDNA. The amplified cDNA products are then detected by DNA analysis techniques.

**Fig. 1.2** below shows a pedigree chart of a family with the disease and their corresponding band patterns.



**Fig 1.2**

(a) Suggest how the products from reverse transcription are processed before PCR is applied. [1]

- **Digest mRNA with RNase** to separate the mRNA from single stranded cDNA.

(b) With reference to **Fig. 1.2**,

(i) Explain how the band patterns shown in Fig.1 2 are obtained. [4]

- **cDNA of fibrillin gene of different individuals** are obtained ...
- ...and amplified by PCR using 2 **primers complementary to the 3' ends of the FBN1 cDNA**.
- The alleles/cDNA are subjected to **gel electrophoresis** where the cDNA are **separated according to size** (OWTTE). **Shorter fragment/allele move further/faster down the gel (vice versa)**
- The gel is stained with **ethidium bromide/DNA binding dye** and **DNA bands** are viewed under **UV light**.

(ii) explain the mode of inheritance of the Marfan syndrome. [3]

- Marfan syndrome is an **autosomal dominant** disease.
- **One copy** of the **mutated allele** is sufficient to result in Marfan syndrome.
- **All affected individuals I-1, II-2 and II-4 have at least one copy of the 1.2 kb allele,**
- ...while the **unaffected individuals I-2, II-1 and II-3 has two copies of the 1.8 kb allele.**
- It is autosomal/ not sex-linked because **affected I1 and unaffected I2 does not result in all affected daughter.**

(iii) explain the band pattern observed in individual II-4. [3]

- II-4 has **thicker 1.2kb band**,
- hence has **2 copies / homozygous** for the **mutant allele**
- One copy is **inherited from his father / individual I-1.**
- Other copy is a result of a **deletion of 0.6 kb nucleotide sequence/ mutation** of the normal allele inherited from **individual I-2.**

**[15 marks]**

## Question 2

Severe combined immunodeficiency (SCID) is a heritable disease of young children in which the patient suffers severe abnormalities of the immune system. As a result, these patients are susceptible to opportunistic infections.

In the autosomal recessive condition of SCID, the patients are homozygous for the mutated ADA gene on chromosome 20.

(a) Explain why patients that are homozygous for the mutated ADA gene are susceptible to opportunistic infections. [3]

- **Mutated ADA allele produces non-functional ADA protein which cannot break down purines**
- **Hence, purine accumulates in the cells and are toxic to T cells and B cells**
- **..prevent the development of T cells /B cells. This leads to the lack of T-cells and B-cells to fight against opportunistic infection.**

(b) Gene therapy for ADA-deficient severe combined immunodeficiency (SCID) can be carried out by several ex vivo methods presented below.

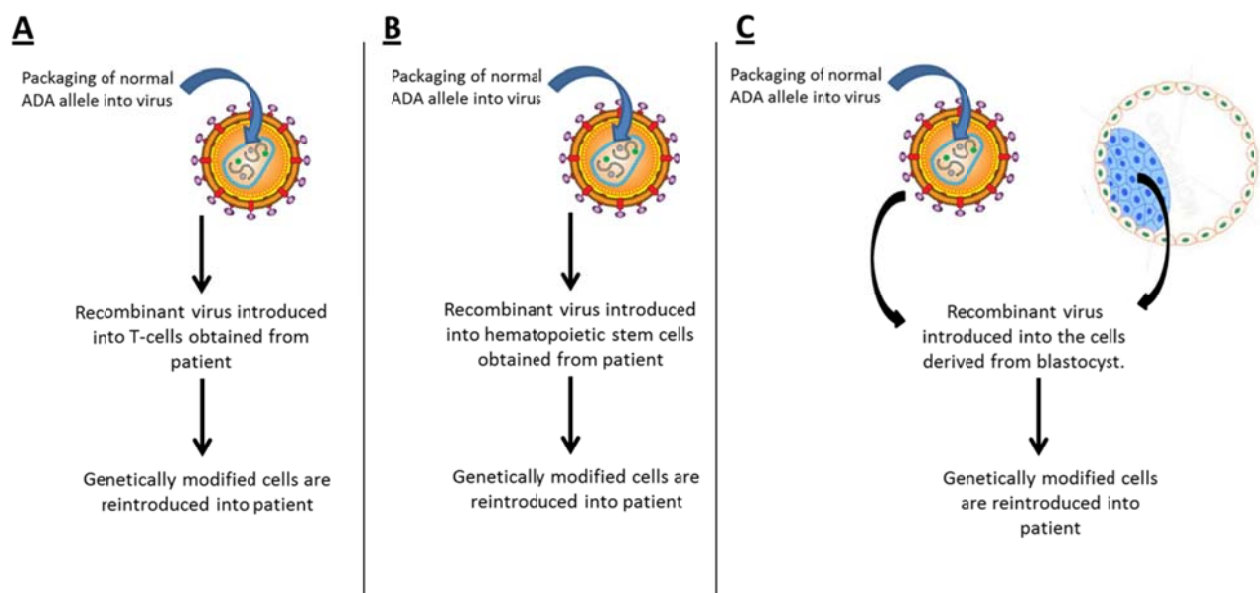


Fig. 2.1

(i) Of the possible gene therapy methods shown above, method B is the most preferred by scientists in treating SCID.

Comment on this preference for method B over the other 2 methods, shown in **Fig. 2.1**. [2]

- **Haematopoietic stem cells (method B) are capable of dividing continually / self-renewal which allow treatment to be more permanent as compared to using T-cells /method A which are differentiated.**
- **Haematopoietic stem cells (method B) are obtained from and returned to patient, hence will not trigger immune response / cause rejection as compared to using embryonic stem cells from another source /method C.**

(ii) Suggest why *ex vivo* approach was adopted for gene therapy treatment. [2]

1. Allows specific **identification of target cells** with therapeutic gene (OWTTE).
2. Allows **monitoring of gene expression before reintroduction** as success rate is very low.
3. Safer because can **monitor for cancer cells** caused by the **random integration of DNA / insertional mutagenesis by retrovirus** which can knock out tumour suppressor genes or activate proto-oncogenes.

(c) Explain why the treatment of cystic fibrosis using stem cells is not suitable. [1]

- Stem cells giving rise to lung epithelial cells are **rare / difficult to obtain**.  
OR  
Modified cells are **difficult to reintroduce** into the lungs of the patient.

Once most stem cells differentiate, they lost their ability to turn into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In an experiment with mice to produce iPS cells, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. These four genes indirectly result in the reprogramming of the differentiated cells by influencing other genes.

(d) Outline how the addition of the four genes may influence the activity of many other genes to result in the reprogramming of cells. [2]

- The **genes code for transcription factors / chromatin remodelling enzymes** that ....

[Give example]

- ...**increases the expression of genes** that control **pluripotency** of cells.

OR

- ...**decreases the expression of genes** that resulted in **differentiation** of cells.

[Total: 10]

### Question 3

Fig. 3.1 shows the proportion of cotton, maize and soybeans grown in the USA that are genetically modified in two different ways:

- HT crops are modified to be resistant to the herbicide glyphosate (originate from xx plant)
- Bt crops are modified to express the Bt toxin which kills insect pests.

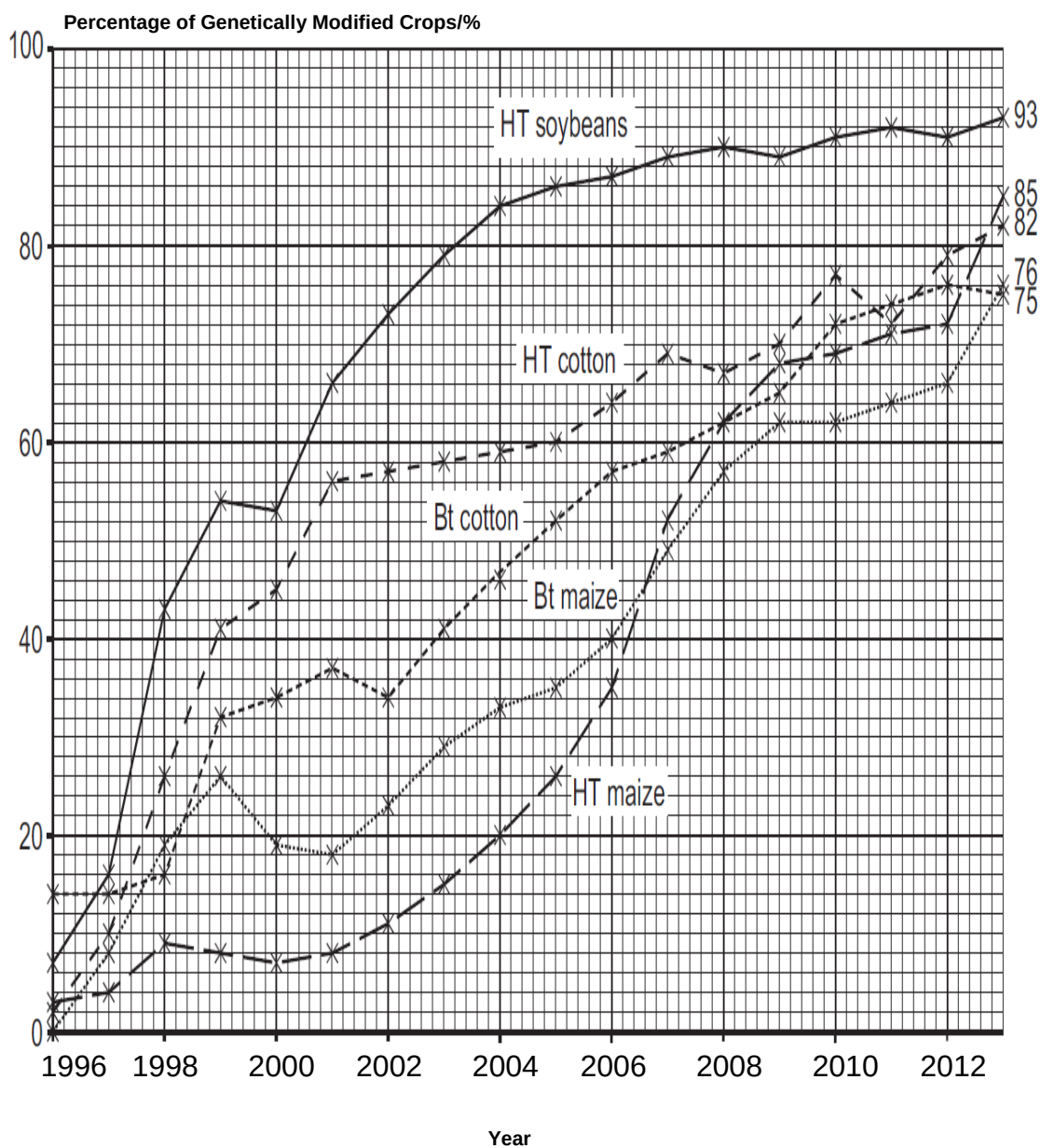


Fig. 3.1

(a) With reference to **Fig. 3.1**,

(i) Describe the change in the percentage of genetically modified cotton grown in the USA since 1996. [2]

- There is an **increase in percentage of genetically modified cotton** grown in USA.
- **Bt cotton** increased from **16%** (in 1996) to **75% in 2013**  
OR  
**HT cotton** increased from **2%** (in 1996) to **82% in 2013**

(ii) Outline how a recombinant vector containing the herbicide glyphosate resistance gene can be produced. [4]

- **Isolate Gene-of-Interest:**
  - 1) cut **gene** coding for glyphosate resistant **from genome using restriction enzyme**
  - 2) Synthesize double-stranded **cDNA** of coding for glyphosate resistant **using reverse transcriptase**.
- **Preparation of complementary sticky ends:**
  - 1) The **same restriction enzyme** is used to cleave the **plasmid and gene** to produce complementary **sticky ends**
  - 2) Add **DNA adaptors/artificial sticky ends** to the blunt ends of **plasmid and gene**.
- **Mix** the cut plasmid and gene for the **complementary sticky ends** to anneal via **hydrogen bonds**.
- **DNA ligase** is added to allow formation of **phosphodiester bonds** between the nucleotides of the gene and plasmid to produce recombinant vector.

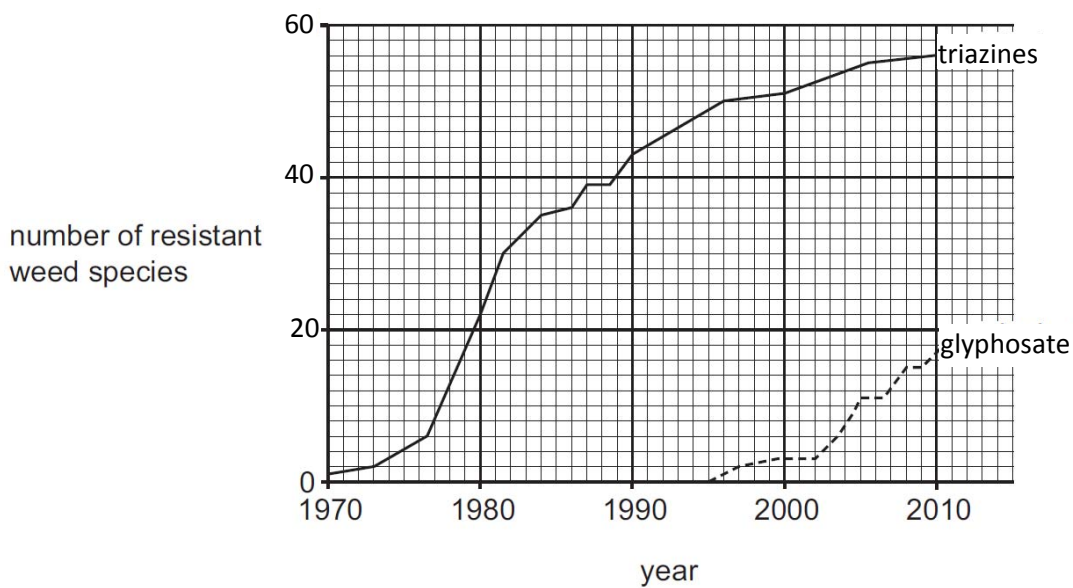
(b) The HT crops received two new genes that gave resistance to glyphosate herbicide and also a marker gene called GUS. The parts of the plant that express the GUS gene turn blue when dipped into a colourless chemical substrate.

Explain why the GUS gene was also transferred to the genetically modified crops. [2]

- Use as a genetic marker to determine whether the **resistance genes** have been **successfully introduced** into the **plant cells**.
- The appearance of the blue **colour** allows **easier identification of genetically modified plant cells**
- It is **easier to identify compared to antibiotic resistance markers**.
- To **check all parts of the plants express the resistance gene** (OWTTE).

[Any 2]

**Fig. 3.2** shows the increase in the number of weed species resistant to glyphosate herbicide and triazine herbicides since 1970. Crops have not been genetically modified to resist triazine herbicides.



**Fig. 3.2**

(c) With reference to **Fig. 3.1** and **Fig. 3.2**, justify whether the rise of herbicide resistant weeds was due to gene transfer from genetically modified maize. [4]

- Glyphosate resistant weeds species is **due to gene transfer** because of the **corresponding increase** in number of **resistance weeds** species and percentage of **HT crops**
- The glyphosate resistance weeds species starts to increase **from 0 in 1995 / 1996 to 17/18 in 2010** (data citation mark for in Fig. 3.2), this correspond (to Fig. 3.1) to the increase in percentage from **3% to 85% in HT maize OR 2% to 82% in HT cotton OR 7% to 93% in HT soybeans in 2013** (Accept 2010 data)
- Triazine-resistance weeds species is **not due to gene transfer** as there were **no genetically modified crops with triazine-resistance** genes in Fig. 3.1.
- **The increase in triazine resistance weeds species occurred before 1996 from 1 to 50.**

(d) A relative of maize, called teosinte, grows in the wild. There are concerns that pollens from Bt maize could pollinate wild teosinte and transfer Bt gene to it.

It was found that the length of time that maize pollen is in the air affects its ability to bring about fertilisation in a female flower. The maximum time a pollen retain its ability to fertilise a female flower while exposed to air is 120 mins.

- (i) Suggest how this information could be used to devise a strategy that would reduce the possibility of transferring Bt gene from Bt maize to the wild teosinte. [1]

- **Determine the distance the pollen can travel in 120 mins and plant the crops further** than that (OWTTE).
- (ii) Suggest how transfer of new genes, such as Bt gene, to teosinte may endanger the environment. [2]
- The **introduced transgenes to teosinte** may result in **more invasive hybrid offspring** / teosinte may **outcompete other species**  
OR  
**Risk of toxins** produced by teosinte **affecting non-target organisms**.
  - **upsetting the natural balance of the ecological system.**

**[Total: 15]**

## Question 4 Planning

Enzymatic reactions are influenced by various factors, such as the presence of inhibitors. There are two main types of inhibitors, the competitive inhibitors and non-competitive inhibitors. The mode of action of the two types of inhibitors differs and both have a different effect on enzyme-catalysed reactions.

It was known that the action of amylase is inhibited by the presence of ions such as copper ( $\text{Cu}^{2+}$ ). Plan an investigation to determine if  $\text{Cu}^{2+}$  functions as a competitive inhibitor or a non-competitive inhibitor of amylase.

Your planning must be based on the assumption that you have been provided with the following apparatus and materials, as well as other appropriate apparatus and equipment found in the laboratory.

- 1  $\text{mol dm}^{-3}$  amylase enzyme
- 5  $\text{g dm}^{-3}$  starch solution
- Benedict's reagent
- 5  $\text{mol dm}^{-3}$  Copper Sulphate ( $\text{CuSO}_4$ ) stock solution
- 10  $\text{mol dm}^{-3}$  maltose solution
- Glass rods
- White tiles
- Bunsen burner, tripod and wire gauze
- Stopwatch
- Access to water bath
- Other measuring instruments (measuring cylinder, syringes, etc) and apparatus (beakers, test-tube, etc)

Your plan should have a clear and helpful structure to include:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it, including scientific reasoning behind procedure.
- Be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- Include the layout of result tables and graphs with clear headings and labels.
- Include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative and precise and reliable as possible.
- Include reference to safety measures to minimise any risks associated with the proposed experiment.

**[Total: 12]**

## Methods (suggested ans)

### **Preparation of maltose colour standards**

1. Prepare 5 different concentrations of maltose solutions according to the dilution table shown below. The volume of each concentration will be the same at  $10\text{cm}^3$ .

| Concentration of maltose /% | Volume of stock maltose solution / $\text{cm}^3$ | Volume of distilled water / $\text{cm}^3$ |
|-----------------------------|--------------------------------------------------|-------------------------------------------|
| 10.0                        | 10.0                                             | 0.0                                       |
| 8.0                         | 8.0                                              | 2.0                                       |
| 6.0                         | 6.0                                              | 4.0                                       |
| 4.0                         | 4.0                                              | 6.0                                       |
| 2.0                         | 2.0                                              | 8.0                                       |
| 0.0                         | 0.0                                              | 10.0                                      |

2. Prepare a boiling water bath using the beaker and Bunsen burner provided.
3. To  $2\text{ cm}^3$  of  $10\text{mol dm}^{-3}$  maltose, add  $2\text{ cm}^3$  of benedict's solution. Mix well by stirring with a glass rod.
4. Repeat step 4 for all other maltose concentration.
5. Place all the test-tubes into the boiling water bath for 2 min.
6. After 2 min, remove all the test tubes, note the colour change and place them in a test-tube rack. The tubes can be used as color standards.

OR

After 2 min, remove the test-tubes. Stir each suspension and  $\frac{3}{4}$  fill a cuvette with each suspension and measure absorbance using a colorimeter. Draw a standard graph.

### **Preparation of different concentration of starch**

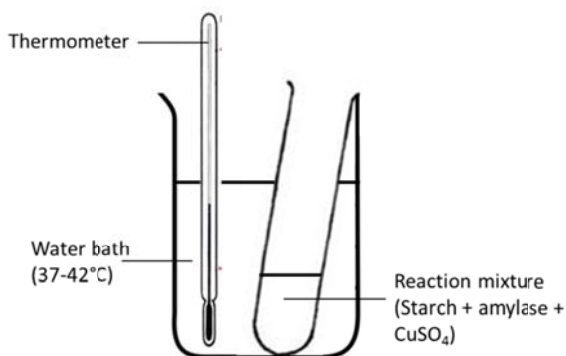
1. Prepare 5 different concentrations of starch according to the dilution table below. Make up the total volume to  $10\text{cm}^3$ .

| Concentration of starch / $\text{g dm}^{-3}$ | Volume of stock starch solution / $\text{cm}^3$ | Volume of distilled water / $\text{cm}^3$ |
|----------------------------------------------|-------------------------------------------------|-------------------------------------------|
| 5.0                                          | 10.0                                            | 0.0                                       |
| 4.0                                          | 8.0                                             | 2.0                                       |
| 3.0                                          | 6.0                                             | 4.0                                       |
| 2.0                                          | 4.0                                             | 6.0                                       |
| 1.0                                          | 2.0                                             | 8.0                                       |
| 0.0                                          | 0.0                                             | 10.0                                      |

2. Acclimatize the amylase solutions and starch suspension in the constant  $38-42^\circ\text{C}$  water bath for 2 min.
3. After 2 min, add  $1\text{cm}^3$  of the  $5\%$   $\text{CuSO}_4$  solution and  $1\text{cm}^3$  amylase solution to  $5\text{cm}^3$  starch solution.
4. Mix well by stirring and start time. Allow the reaction to take place for 1 minute.
5. Immediately after 1 minute, remove the reaction mixture.
6. To  $2\text{cm}^3$  of the reaction mixture, add equal volume of benedict's solution and carry out Benedict's test.
7. Compare the colour change with the colour standard /standard graph prepared earlier to determine the concentration of maltose produced.
8. Repeat steps 2 to 9 using fresh solutions to obtain two further readings.

9. Repeat steps 2 to 10 using each concentration of starch solution.
10. Repeat steps 2 to 11 / repeat the entire experiment, but do not add  $\text{CuSO}_4$  solution this time. This is to obtain results of the effect of [starch] on the rate of starch hydrolysis / enzyme activity on starch hydrolysis without the presence of inhibitor.
11. **Control:** Replace amylase with distilled water. To show that any production of glucose still due to the presence of amylase which hydrolyses starch, and hence the results obtained with and without  $\text{CuSO}_4$  is due to the  $\text{CuSO}_4$  effect on amylase.

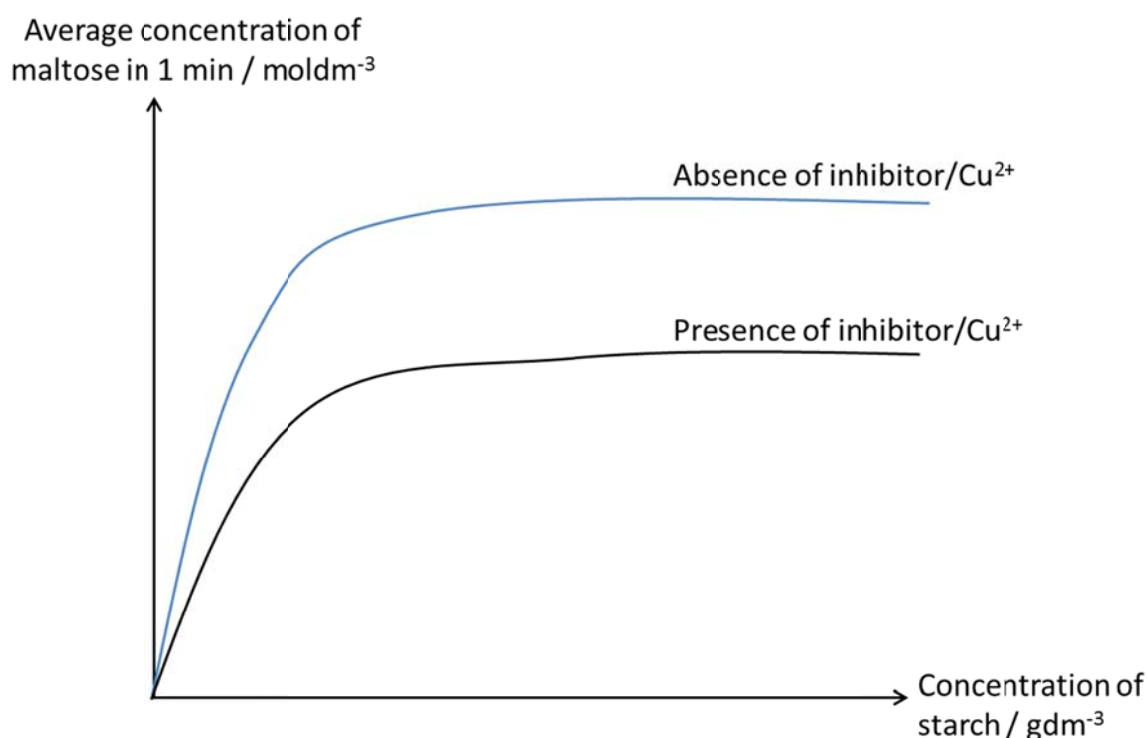
**Diagram [1]**



**Results – Table [1 for correct proposed layout]**

| Concentration of starch / $\text{gdm}^{-3}$ | Concentration of maltose concentration produced in the <b>presence</b> of $\text{Cu}^{2+}$ in 1 min/ $\text{mol dm}^{-3}$ |    |    |         | Concentration of maltose concentration produced in the <b>absence</b> of $\text{Cu}^{2+}$ in 1 min/ $\text{mol dm}^{-3}$ |    |    |         |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----|----|---------|--------------------------------------------------------------------------------------------------------------------------|----|----|---------|
|                                             | R1                                                                                                                        | R2 | R3 | Average | R1                                                                                                                       | R2 | R3 | Average |
| 5.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 4.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 3.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 2.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 1.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |
| 0.0                                         |                                                                                                                           |    |    |         |                                                                                                                          |    |    |         |

## Results – Graph [1 for correct proposed graphS]



Copper ion functions as a non-competitive inhibitor by binding to a site away from the active site. The binding alters the tertiary structure of the enzyme and prevents the rate of reaction from reaching  $V_{max}$ .

## Risk assessment [1]

| Risk                                                         | Precaution                                                                                              |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Copper sulfate / amylase / starch is an irritant to the skin | Wear gloves and goggles when handling / Wash hands thoroughly if hands come into contact with solution. |
| Boiling water may scald/burn skin                            | Use a test-tube holder to hold test-tube. Care must be taken when handling the boiling water.           |

## Question 5

(a) Describe the elongation process, including the roles of primers, in PCR. [6]

### Elongation

1. At **72°C**, the optimal temperature for **Taq polymerase**.
2. **Taq polymerase** adds deoxyribonucleotides to growing daughter strand in the **5' to 3' direction**.
3. by **complementary base pairing**.
4. catalyses **formation of phosphodiester bond between adjacent nucleotides**.

5. Each cycle is repeated 30-40 times.
6. The **amount of DNA is doubled after every cycle / Exponential increase** of target DNA sequences as product.

### **Roles of primers**

7. Primers anneal to **3' ends flanking the target DNA sequence** provide the **3'OH group** for the **addition of deoxyribonucleotides by Taq polymerase**
8. Primers anneal to **3' ends flanking the target DNA sequence** to **amplify only** the specific **target** sequence
9. **2 different primers** (forward and reverse primers) needed as the **sequences of the 2 DNA strands are different / not palindromic**.
10. **Excess primers** present **prevent the denatured DNA strands/template from reannealing**.

- (b) Outline the steps in obtaining a callus culture and explain the uses of callus culture in various tissue culture techniques. [8]

### **Obtaining callus culture**

1. **Surface-sterilize** explants / meristematic regions of explants with dilute **sodium hypochlorite solution**
2. Cut explants **into small fragments / wound the explant**.
3. Transfer to culture vessels containing **nutrient agar media** and **cytokinins/plant growth regulators** (intermediate ratio auxin :cytokinin) for **inducing callus** formation by mitosis
4. **Seal and incubate in sterile containers / aseptic conditions**
5. Callus is a **mass of rapidly dividing, undifferentiated/unspecialized cells** which are **totipotent**.

### **Uses of callus culture**

6. Callus is **induced to differentiate** into different **cell or tissue types** with the use of **plant growth regulators**.
7. Number of callus cultures **increased by subculture**.
8. **Sub-cultured callus** can be induced by adjusting the **auxin and cytokinin ratios** to produce **roots** (high auxin: cytokinin) and then **shoots** (low auxin: cytokinin), which eventually **give plantlets / for micropropagation**.
9. **Haploid callus** formed from **anther culture** allow for **selection of characteristics** (eg. Herbicide resistance) to form **diploid callus**.
10. Calli are also useful to **make protoplasts** using cellulase to make the cell walls are thinner than other plant cells, **to insert foreign DNA**.

11. Callus suspensions can be **induced to form somatic embryos**. This can be done by adding auxin and reducing nitrogen levels.
12. Callus can be **genetically modified by introducing beneficial / desired genes** to eventually give **transgenic plant that are insect / pest/ disease resistant**
13. Propagates plants that are depends limited by seasons **throughout the year / produced continuously** to meet consumer demands.
14. For micropropagation of **plant species that are difficult to propagate by traditional methods**. For example, bananas and orchids which have very low seed germination rates.
15. Obtain **whole plants without sexual reproduction**.
16. Obtain **genetically identical plants in large numbers**.
17. Grow disease-free plants / plants **free from disease caused by bacteria and viruses** as callus **obtained from meristems (uninfected)**.

(c) Discuss the ethical concerns that arose from the human genome project. [6]

Ethical issues arises

1. on the **fairness in the use of genetic information** by insurers or other organisations, etc which may create a situation in which **less healthy individuals / those with genetic predisposition are marginalised or discriminated**.
2. on the **confidentiality and privacy of genetic information** - the **right of ownership** of the genetic information has **yet to be determined**.
3. on the use of **genetic information** in **controversial decisions** in reproduction, such as **modifying genotypes through gene therapy**. Hence resulting in the ethical issue of
  - a) one generation **determining the next generation's genotype before birth without their consent**.
  - b) going **against nature** by **tampering** with the next generation's **genetic make up**.
4. on the use of **genetic information** in **controversial decisions** in reproduction, such as **termination of pregnancies**.
  - a) Issue - **selective breeding / abortion based on genetic tests that may not be fully reliable/understood**. → increased abortion rate / decline in appreciation of the dignity of life
5. from **clinical issues** that the doctors, other health service providers, patients and the public must be **educated** to make **informed choices**, and **be aware of scientific capabilities and limitations**.

OR

Unethical for doctors or health care providers to **carry out genetic tests**

- a) **and draw conclusions** on data that may **not be fully reliable** ;
- b) **without informing** the patient of the **limitations and capabilities of genetic technology** ;

- c) leaving patient to interpret complex results without sufficient explanation.
- 6. from commercialization/patenting of genetic information and their product by companies which may limit their accessibility and development of useful medicinal products.
- 7. from **the conceptual and philosophical understanding** of human responsibility, free will vs genetic determinism, and concepts of health and disease.
  - a. E.g. Difficult to determine if a person's behaviour is genetically determined or can be controlled by the individual.

• End of Paper •